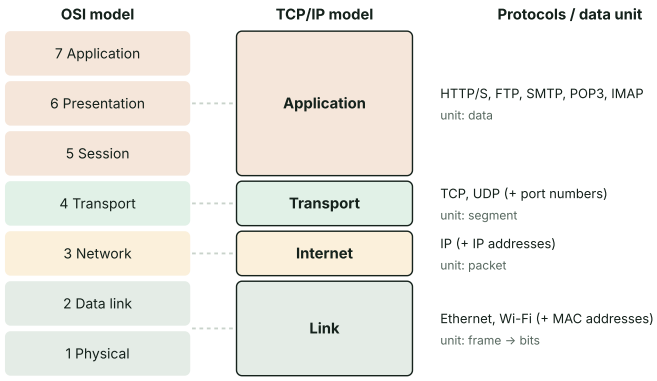
Layered network models and protocols

IS10 FACT FILE

Specification: students should be able to

- Explain why layered models are used in networking
- Describe the layers of the OSI and TCP/IP models
- Describe the purpose of common protocols: HTTP/HTTPS, FTP, SMTP, POP3, IMAP, TCP, UDP, IP, Ethernet and Wi-Fi

Key words: protocol, layer, OSI model, TCP/IP model, encapsulation, HTTP, HTTPS, FTP, SMTP, IMAP, TCP, UDP, IP, Ethernet



The seven OSI layers mapped to the four TCP/IP layers, with example protocols and the header added at each layer during encapsulation

Why use layers

Networking is divided into layers so that each deals with one part of the problem. Each layer offers services to the one above and relies on the one below, communicating through standard interfaces. This means one layer can be replaced, such as wired Ethernet with Wi-Fi, without changing the others. Layers make networks easier to design, develop, teach and troubleshoot, and open standards let hardware and software from different manufacturers work together.

OSI and TCP/IP

The seven-layer OSI model is a reference model: physical, data link, network, transport, session, presentation and application. The four-layer TCP/IP model is the practical model used on the internet: link, internet, transport and application, with the top three OSI layers combined into the application layer. Data is encapsulated on the way down, each layer adding a header, and de-encapsulated on the way up at the receiver.

Key protocols

HTTP transfers web pages and HTTPS does the same over an encrypted TLS connection. FTP transfers files. SMTP sends email, while POP3 downloads it to one device and IMAP keeps it on the server synchronised across devices. TCP provides reliable, ordered, connection-oriented delivery; UDP provides fast, connectionless delivery without guarantees. IP addresses and routes packets. Ethernet and Wi-Fi handle transmission across wired and wireless links.

T Key terms from the mark schemes

Protocol
A set of rules that governs how data is formatted, transmitted and received so devices can communicate.

OSI model
A seven-layer reference model of network communication: physical, data link, network, transport, session, presentation and application.

TCP/IP model
The four-layer model used by the internet: link, internet, transport and application.

Encapsulation (networking)
The process by which each layer adds its own header to data as it passes down the layers before transmission.

TCP
Transmission Control Protocol: a connection-oriented transport protocol giving reliable, ordered delivery using acknowledgements and retransmission.

UDP
User Datagram Protocol: a connectionless transport protocol that is fast but does not guarantee delivery or order.

Exam focus

When asked for an advantage of layering, explain the consequence: 'a layer can be changed without affecting other layers, so a new transmission medium can be introduced without rewriting applications'. When describing a protocol, state its purpose precisely rather than just expanding the acronym. For TCP versus UDP choices, justify using the application: lost data in a file download is unacceptable (TCP) but a late video frame is useless anyway (UDP). Learn which layer each protocol belongs to.

? Check yourself

1. Define the term protocol. [1]
2. Explain two advantages of using a layered model for network communication. [4]
3. Name the four layers of the TCP/IP model in order from the top. [2]

Answers: 1. A protocol is a set of rules that governs how data is formatted, transmitted and received between devices (1). 2. Any two advantages, each stated (1) and explained (1): one layer can be changed or replaced without affecting the others (1) / e.g. Wi-Fi can replace Ethernet at the link layer without changing TCP or HTTP (1) / complex communication is split into smaller, manageable tasks (1) / so each layer is easier to design, develop, teach and troubleshoot (1) / layers follow open standards (1) / so equipment and software from different manufacturers can interoperate (1). 3. Application, Transport (1) / Internet (accept Network), Link (accept Network access / Network interface) (1). All four in the correct order from the top for both marks; award 1 mark if all four are named but the order is wrong.